

Experiment 05

LAN Installation and Configuration

Objective:

To **install and configure a Local Area Network (LAN)** so that multiple computers can communicate and share resources within a limited area like a computer lab or classroom.

Prerequisites:

Before performing this experiment, students should have:

- Basic knowledge of **computer networking**.
- Understanding of **network devices** like switches, routers, cables, NIC.
- Awareness of **IP addressing** (IPv4).

What is LAN?

A **Local Area Network (LAN)** is a network that connects computers and devices in a **limited geographical area**, such as a room, building, or campus. It allows devices to **share resources** like files, printers, and internet connection.

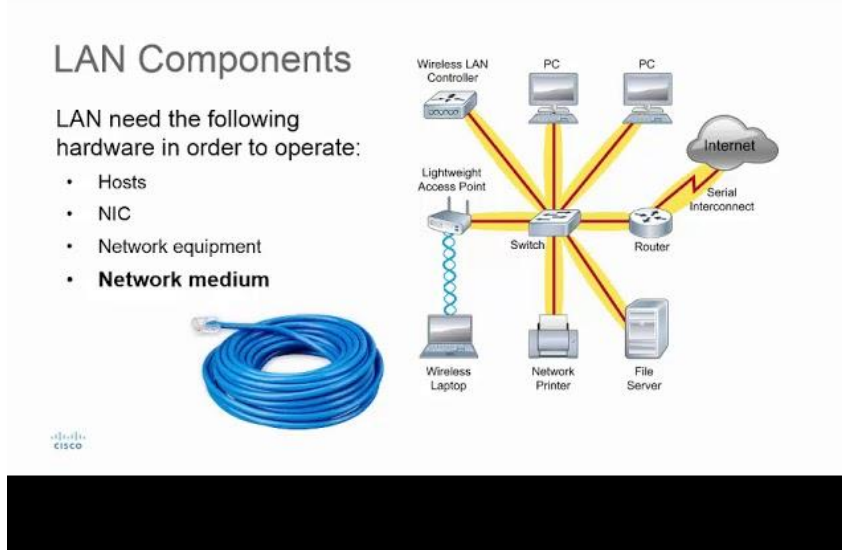
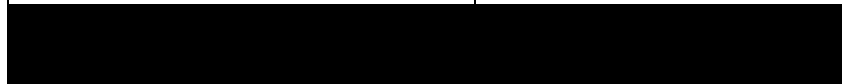
Key Features of LAN:

- Covers small area (home, lab, office).
- High-speed communication (up to 1 Gbps).
- Allows file, printer, and internet sharing.
- Secure and easy to maintain.

Common LAN Devices:

Device	Function
NIC (Network Interface Card)	Allows computer to connect to the LAN

Switch	Connects multiple devices in the LAN
Router	Connects LAN to the internet (if needed)
Ethernet Cable	Physical medium for connection (Cat5e/Cat6)
RJ45 Connector	Connector at the end of Ethernet cables



Apparatus Required:

- 2 or more Computers / Laptops
- Network Switch or Router
- Ethernet (LAN) Cables – Cat5e/Cat6
- RJ45 connectors
- Crimping Tool (optional)
- Power Source

Procedure:

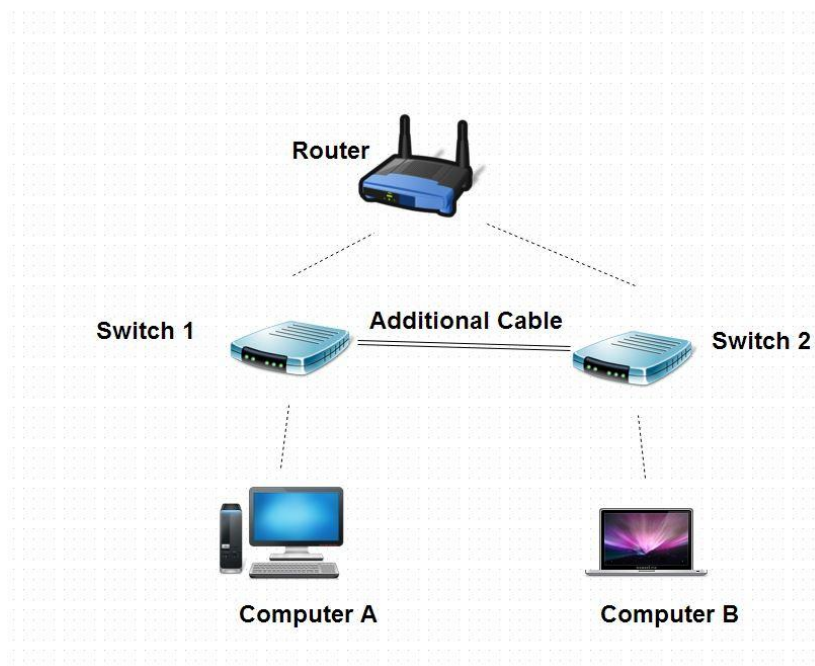
Step 1: Setup Devices

- Place computers close to a **switch or router**.
- Plug the power supply for all devices.

- Check if all computers have **working LAN ports** and **NIC drivers installed**.

Step 2: Make the Connections

- Connect each computer to the **switch/router** using **Ethernet cables**.
- Ensure all cables are tightly plugged into the **RJ45 port**.
- Power on all devices.



Step 3: Configure IP Addresses

Method 1: Static IP

1. Go to: Control Panel → Network and Sharing Center
2. Click: Change adapter settings → Right-click on Ethernet → Properties
3. Select: Internet Protocol Version 4 (TCP/IPv4)
4. Choose: "Use the following IP address"
 - IP Address: 192.168.1.X (X = 1,2,3...)
 - Subnet Mask: 255.255.255.0
 - Default Gateway: 192.168.1.1

♦ Method 2: DHCP (Automatic IP)

- If using a router, enable **DHCP**.
- Computers will automatically receive IP addresses.

Step 4: Test LAN Connectivity

- Open **Command Prompt** on one computer.
- Type: ping 192.168.1.2 (another PC's IP)
- If reply is received, LAN is working.

Reply from 192.168.1.2: bytes=32 time<1ms TTL=128

```
C:\>ping 192.168.1.100

Pinging 192.168.1.100 with 32 bytes of data:

Reply from 192.168.1.100: bytes=32 time<1ms TTL=128
Reply from 192.168.1.100: bytes=32 time<1ms TTL=128
Reply from 192.168.1.100: bytes=32 time<1ms TTL=128
Reply from 192.168.1.100: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.1.100:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>
```

Step 5: File and Folder Sharing (Optional)

- Right-click any folder → Properties → Sharing → Share.
- Ensure network discovery and file sharing are enabled:
 - Go to: Control Panel → Network and Sharing Center → Advanced sharing settings.

Result:

Successfully configured a **LAN network** that enables communication between connected computers.

Conclusion:

From this experiment, we learned:

- How to **set up a LAN** using switch/router and Ethernet cables.
- How to assign IP addresses using **static or DHCP methods**.
- How to **test network connectivity** using the **ping command**.
- Basic resource sharing in a LAN.

Advantages of LAN:

Advantage	Explanation
High Speed	Fast data transmission within local area
Resource Sharing	Devices can share printers, files, internet
Low Cost	Economical setup and maintenance
Secure	Easier to manage and protect from outside access